

BRCA-associated protein1 (BAP1) immunohistochemical stain reliability in postbrachytherapy uveal melanoma enucleation specimens

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Purpose: The BRCA-associated protein1 (BAP1) immunohistochemical (IHC) stain has emerged as a powerful and inexpensive prognostic tool in uveal melanoma (UM), correlating with UM genetics and outcome. The data on the reliability of BAP1 immunohistochemistry in previously irradiated UM is scant. We aim to assess BAP1 IHC in post-Iodine-125 plaque brachytherapy-treated UM-enucleated eyes. **Methods:** In a case-control study, the medical records of all patients who underwent enucleation for UM at a major Ocular Oncology Service from December 1st, 2007 to December 31st, 2014 were reviewed. All cases with either chromosome 3 (ch3) status or sufficient follow-up (>5 years or metastasis) were selected. Nuclear BAP1 (nBAP1) immunoreactivity was interpreted as intact (positive in >90% of nuclei), lost (positive in <5% of nuclei), or heterogeneous (positive in 5–90% of nuclei). Retina and intratumoral blood vessels served as internal positive controls. **Results:** A comparison of 34 postbrachytherapy UM secondary-enucleated eyes with 47 nonbrachytherapy primary enucleated controls revealed no significant difference with respect to nBAP1 IHC (lost in 41% vs 51%, $P = 0.19$), ch3 status (ch3 monosomy in 59% vs 60%, $P = 0.48$), and outcome (metastatic disease in 44% vs 47%, $P = 0.8$). Association of nBAP1 IHC with ch3 status and outcome [intact nBAP1/ch3 disomy and/or no metastasis) and lost nBAP1 (ch3 monosomy and/or metastasis)] in post-brachytherapy UM was significantly lower when compared with non-brachytherapy tumors [21/30 (70%) vs 41/44 (93%), $P = 0.004^*$]. **Conclusion:** Although nBAP1 IHC stain is a strong prognostic tool in UM, its association with ch3 status, and outcome in postbrachytherapy UM was significantly lower compared with nonbrachytherapy tumors due to pitfalls in the interpretation of nBAP1 immunoreactivity in irradiated UM. This test should be used judiciously in the prognostication of postbrachytherapy-enucleated UM.

Key words: Brachytherapy, BAP1, chromosome 3, enucleation, immunohistochemistry, metastases, uveal melanoma

Uveal melanoma (UM), the most common primary intraocular malignancy in adults, is associated with up to 50% mortality from metastatic disease.^[1] Thus, accurate prognostication of UM is invaluable in guiding oncologic surveillance and management. The loss of chromosome 3, i.e. monosomy 3 (M3) is a major prognostic factor in UM. The M3 UM, defined by the cancer genome atlas as class C or D,^[1] has a 33% and 63% 5-year metastatic risk, respectively.^[1] Class 2 gene expression profile (GEP) is another major unfavorable prognostic factor in UM, associated with a 5-year 41% metastasis-free survival rate^[2] and with the M3 status.^[3] Inactivating mutations in a tumor-suppressor *BRCA-associated protein1* (*BAP1*) gene on chromosome 3p21.1 were identified in 26/31 (84%) metastasizing UM.^[4] *BAP1* gene inactivation frequently

occurs through the loss of one chromosome 3 allele and an additional *BAP1* mutation on the remaining allele.^[5]

The BAP1 immunohistochemical (IHC) stain has emerged as a valuable inexpensive prognosticator in UM. The loss of the nuclear BAP1 (nBAP1) protein expression by IHC correlates with somatic *BAP1* gene mutations, irrespective of the type of gene mutation.^[5] Additionally, loss of nBAP1 expression correlates with chromosome 3 ploidy.^[6] Tabuenca Del Barrio *et al.*^[7] documented that BAP1 protein expression by IHC is a major prognostic factor of metastasis-free survival. See *et al.*^[8] found that low nBAP1 IHC (staining of less than 33% of tumor cells per high-power field, assessed in 3 high-power fields) also correlates strongly with GEP class 2 and is equally accurate in metastasis prediction. Van de Nes *et al.*^[5] and Patrone *et al.*^[9] found that the loss of nBAP1 protein expression by IHC has a stronger correlation with patient survival than somatic *BAP1* gene mutation status.

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Plaque brachytherapy is the most common treatment modality for small and medium-sized UMs. Although Iodine-125 (I-125) plaque brachytherapy is a highly effective method for local tumor control, enucleation may be required for recurrent tumors or in eyes with treatment-related complications.^[10]

The comparative genetic data on nonirradiated and irradiated UM are mixed. Several studies showed that genetic profiling can be obtained in previously irradiated tumors.^[3,11-13] Hussain *et al.*^[11] documented that in eyes that underwent fine needle aspiration biopsy (FNAB) 0–20 days postplaque brachytherapy, genetic studies were successful in all 102 cases. Onken *et al.*^[3] showed that FNAB can be obtained postradiotherapy in 15 patients and Gold *et al.*^[12] had similar findings in three patients. Thornton *et al.*^[13] compared 172 tumors with preirradiation FNAB to 176 tumors sampled up to 284 days postradiation with comparable genetic results. The concordance of genetic studies between pre- and post-brachytherapy specimens was examined by Coupland *et al.*^[14] who showed that chromosome 3 data did not differ before and after radiotherapy in 4 patients. Wackernagel *et al.*^[15] demonstrated comparable chromosome 3 and 8 statuses in 5 patients. Other investigators noted some differences between the genetic profiles between nonirradiated and irradiated tumors. For example, Matet *et al.*^[16] compared chromosomal status in endoresection specimens in 17 patients and found identical chromosomes 3 and 8 profiles in 94%, with 18% discordance in other chromosomes. Dogrusöz *et al.*^[17] compared 36 irradiated globes to 291 primary enucleated globes and showed that karyotyping and FISH failed significantly more often in irradiated tumors, with more common and complex aberrations.

Similar to molecular genetic testing, immunohistochemical stains can be influenced by irradiation induced necrosis, tissue damage, and ischemia.^[18,19] Irradiation can also induce epigenetic changes that affect protein expression.^[20] The data on the reliability of BAP1 immunohistochemistry in the previously irradiated UM is scant. The study by Szalai *et al.*^[21] included 18 patients with UM previously treated by plaque brachytherapy, proton beam radiotherapy, or ophthalmic artery embolization who underwent secondary enucleation and 22 patients with primary enucleation as treatment of UM. The investigators found a significant difference in nBAP1 protein expression by immunohistochemistry between patients with and without prior therapy, but specific information on the difference between the two groups was not provided.

We conducted this study to assess the reliability of the BAP1 immunohistochemical stain in prognostication of patients with UM, who underwent enucleation post I-125 brachytherapy.

Methods

In this retrospective single-center institutional case-control study, the medical records of all patients who underwent enucleation for uveal melanoma at a major Ocular Oncology Service between December 1st, 2007 and December 31st, 2014 were reviewed for clinical, histopathologic, and genetic data. All cases with available paraffin-embedded tissue and either known chromosome 3 status or sufficient follow-up (development of metastasis or follow-up > 5 years) were selected for analysis. The study followed the tenets of the

Declaration of Helsinki and was approved by the Institutional Review Board.

Clinical information

Clinical data collected included patient age, sex, race, symptoms, clinical characteristics of the lesion (location, size, clinical American Joint Committee on Cancer [AJCC] stage^[22]), primary treatment, FNAB route, reason for enucleation, follow-up duration, and outcome.

Pathologic evaluation

Routine sections stained with hematoxylin-eosin were prepared from paraffin-embedded, formalin-fixed tissues. Immunohistochemical staining for BAP1 (clone sc-28383, 1: 40 dilution; Santa Cruz Biotechnology, Dallas, TX, USA) was performed on 4- μ m-thick tissue sections with an automated IHC staining system (Leica Bond-III automated system; Leica Microsystems, Chicago, IL, USA) with appropriate positive and negative controls in accordance with the manufacturer's guidelines as previously described^[8] and detected with 3,3'-Diaminobenzidine (DAB) chromogen. For initial BAP1 immunohistochemical stain validation in UM, we used two paraffin-embedded enucleated globes with untreated UM (BAP1-wild type, BAP1 IHC-intact UM, and BAP1-mutated, BAP1 IHC-lost UM) from the study by See *et al.*^[8] The pathology material was reviewed by two observers and a consensus interpretation of histopathology and immunohistochemistry was rendered. The histopathologic parameters assessed included tumor size, involved structures, growth pattern, mitotic activity, percentage of tumor necrosis, and pathologic AJCC stage and grade.^[22] BAP1 immunoreactivity was interpreted as intact (positive in >90% of nuclei), lost (positive in <5% of nuclei), or heterogeneous (positive in 5–90% of nuclei). Retina (when available) or intratumoral inflammatory cells and blood vessels served as internal positive controls. Eyes without sufficient internal positive controls or with <5% viable tumor were labeled "noninterpretable".

Genetic information

The information on BAP1 mutation status and chromosome 3 status was extracted from the medical records, when available. Samples for genetic analysis were obtained using FNAB, and the genetic studies were performed at the Genetic Diagnostic Laboratory at the University of Pennsylvania as previously described.^[1] Chromosome 3 and 8 analyses of UM were performed in 22/34 (65%) patients who had secondary enucleation. Of these 22 UMs, genetic testing was performed on nonbrachytherapy tumors in 16 (73%) cases (sample obtained at the time of plaque brachytherapy placement) and following brachytherapy in 5 (23%) cases (sample obtained at the time of enucleation). One (4%) patient had genetic testing of their tumor performed twice, before brachytherapy and at the time of enucleation (which had similar results). A UM sample for genetic studies was obtained at the time of enucleation in 47/47 (100%) patients who had primary enucleation.

Statistical analysis

The association between immunoreactivity for BAP1, chromosome 3 status, and metastatic disease development were assessed in patients who underwent primary enucleation and compared to patients who underwent enucleation following

plaque brachytherapy. We performed a stratified analysis, in which we subdivided our primary UM and postbrachytherapy UM groups. We then tested for binomial proportions between the specific subgroups comparing the primary and postbrachytherapy groups. The *P* value for binomial proportions was calculated. A *P* value < 0.05 was considered statistically significant.

Results

Records search identified 152 paraffin-embedded tissue blocks from 152 patients with UM, who underwent enucleation between December 1st, 2007 and December 31st, 2014. Eighty one of 152 patients met study inclusion criteria, yielding 34 patients (34 eyes), who underwent secondary enucleation following I-125 plaque brachytherapy for UM and 47 patients (47 eyes) with untreated tumors, who underwent primary enucleation (controls). Seventy-one patients, who had no data on chromosome 3 status or sufficient follow-up (development of metastasis or follow-up >5 years) were excluded from the analysis.

Clinical information

Clinical features in both cohorts on initial assessment are listed in Table 1. The age, sex, and race of patients with postbrachytherapy UM and nonbrachytherapy UM were similar.

Although patients with postbrachytherapy tumors had less frequent symptoms of blurred vision (59% vs. 81%, *P* = 0.03), there was no significant difference in visual acuity (visual acuity 20/50–20/400 in 35% vs. 47%, *P* = 0.30) [Table 1]. Clinically, the postbrachytherapy tumors were significantly smaller in their largest basal diameter (14 mm vs. 17 mm, *P* < 0.001) and thickness (6 mm vs. 11 mm, *P* < 0.001) with less frequent ciliary body involvement (10/32 vs. 29/46, *P* = 0.006), more frequent AJCC cT1 category tumors (9 vs. 2, *P* = 0.004) and less frequent AJCC cT4 category tumors (5 vs. 27, *P* < 0.001) [Table 1].

Pathologic evaluation

The histopathologic features in both cohorts are listed in Table 2. The post-brachytherapy tumors were thinner (5 mm vs 9 mm, *P* < 0.001) and had less frequent mushroom-shaped configuration (24% vs. 49%, *P* = 0.02), fewer mitotic figures (<10 mitotic figures in 40 high-powered-fields in 59% vs. 34%, *P* = 0.03), greater frequency of necrosis (68% vs. 28%, *P* < 0.001), and fewer AJCC pT3 tumors (9 vs. 29, *P* = 0.002) [Table 2]. There was no difference between post-brachytherapy and nonbrachytherapy UM with respect to other histopathologic parameters [Table 2].

The immunohistochemical and molecular genetic features in both cohorts are listed in Table 3. BAP1 IHC was noninterpretable in 7/81 (9%) eyes (4 postbrachytherapy, 12% and 3 nonbrachytherapy, 6%, *P* = 0.2). There were no cases with heterogeneous BAP1 IHC. Therefore, we regarded the BAP1 IHC in a binary fashion, as has been cited in previous studies.^[6] There was no significant difference between the postbrachytherapy and non-brachytherapy tumors with respect to nBAP1 IHC status (lost in 41% vs. 51%, *P* = 0.19) or chromosome 3 status (ch3M is 59% vs. 60%, *P* = 0.48) [Table 3].

Follow-up data and date of last seen clinical features in both cohorts are listed in Table 4. The follow-up of patients with

postbrachytherapy tumors was significantly longer (90 months vs. 52 months, *P* < 0.001). In 64 patients with available follow-up information of more than 60 months, there was no significant difference between those with postbrachytherapy and nonbrachytherapy UM with respect to local recurrence (7% vs. 3%, *P* = 0.55), metastatic disease (44% vs. 47%, *P* = 0.8), and disease-related mortality (31% vs. 31%, *P* = 1) [Table 4].

Genetic information

There were 8 of 81 patients (10%) including 5 with nonbrachytherapy UM and 3 with post-brachytherapy UM, who were tested for *BAP1* mutation. Three of 5 (60%) nonbrachytherapy UM had somatic *BAP1* mutation (*BAP1*^{MUT}) correlating with the loss of nBAP1 expression in all 3 tumors. Out of 2 non-brachytherapy UM with wild-type *BAP1* (*BAP1*^{WT}), one had intact nBAP1 expression, and one had lost nBAP1 expression (he had monosomy 3 and developed metastatic disease). Of three patients with postbrachytherapy UM, one had germline *BAP1*^{MUT} correlating with the loss of nBAP1 expression in the tumor, one had *BAP1*^{WT} correlating with intact nBAP1, and one had somatic *BAP1*^{MUT} with a non-interpretable BAP1 IHC. The small sample size precludes the evaluation of statistical significance of these findings.

Of 11 primary UM with intact nBAP1 and available follow-up data, 9 (82%) had no metastases. Of 15 post-brachytherapy UM with intact nBAP1 and available follow-up data, 9 (60%) had no metastases. The difference was not statistically significant (*P* = 0.18). Of 19 primary UM with lost nBAP1 and available follow-up data, 13 (68%) developed metastases. Of 13 postbrachytherapy UM with lost nBAP1 and available follow-up data, 7 (54%) developed metastases (*P* = 0.20).

The association between nBAP1 expression, chromosome 3 status, and outcome is summarized in Table 3 and illustrated in Figs. 1–3.

Primary UM

Concordant cases: 17 of 20 (85%) tumors with intact nBAP1 were concordant with respect to either genetic status (16 tumors with ch3D) or outcome (1 patient with ch3M and no metastasis at 5 years) [Fig. 1a, Table 3]. All 24 primary UM with lost nBAP1 had either ch3M (24/24, 100%) or metastasized (13/19, 68%) [Fig. 1b and c, Table 3].

Discordant cases: Five of 20 (25%) UM patients with intact nBAP1 were discordant with respect to either genetic status (4 tumors with ch3M) [Fig. 2a and b] or outcome (1 patient with metastases and ch3D). The other patient with metastases also had ch3M). Out of 24 patients with lost nBAP1, none had ch3D (0%), and 6 had no metastatic disease (6/19, 32%) [Table 3].

Post-brachytherapy UM

Concordant cases: Eleven of 16 (69%) tumors with intact nBAP1 were concordant with respect to either ch3D status (5 tumors with ch3D) or outcome (6 patients with ch3M or no genetic data and no metastases) [Fig. 1d]. Ten of 14 (71%) irradiated UM with lost nBAP1 had either ch3M (7 tumors) or metastasized (3 patients with ch3D or no genetic data) [Fig. 1e and f, Table 3].

Discordant cases: Seven of 16 (44%) UM patients with intact nBAP1 were discordant with respect to either genetic status (3 tumors with ch3M) or outcome (4 patients with ch3D or no genetic

Table 1: Demographic features and initial assessment characteristics of uveal melanoma in 81 patients comparing primary versus secondary enucleation

	Primary enucleation (%) Non-irradiated melanoma (n=47 patients, 47 eyes)	Secondary enucleation (%) Irradiated melanoma (n=34 patients, 34 eyes)	P*	All patients (%) (n=81 patients, 81 eyes)
Age (months)				
Mean (median, range)	56 (59, 20–91)	57 (56, 23–89)	0.85	56 (58, 20–91)
Sex				
Male, Female	20 (43%), 27 (57%)	18 (53%), 16 (47%)	0.36	38, 43
Race				
Caucasian	46/47 (98%)	32/34 (94%)	0.38	78/81 (96%)
African American	1/47 (2%)	0/34 (0%)		1/81 (1%)
Hispanic	0/47 (0%)	0/34 (0%)		0/81 (0%)
Asian/South Asian	0/47 (0%)	2/34 (6%)		2/81 (2%)
Main symptom				
Asymptomatic	1/47 (2%)	4/34 (12%)	0.08	5/81 (6%)
Blurred vision	38/47 (81%)	20/34 (59%)	0.03	58/81 (72%)
Floaters	4/47 (9%)	1/34 (3%)	0.30	5/81 (6%)
Other	4/47 (9%)	9/34 (26%)	0.03	13/81 (16%)
Visual acuity				
≥20/40	9/47 (19%)	13/34 (38%)	0.06	22/81 (27%)
20/50-20/400	22/47 (47%)	12/34 (35%)	0.30	34/81 (42%)
<20/400	16/47 (34%)	9/34 (26%)	0.46	25/81 (31%)
Tumor size (mm)				
Basal diameter (median, range)	17 (18, 5–24)	14 (15, 4–20)	< 0.001	16 (16, 4–24)
Thickness (median, range)	11 (12, 3–20)	6 (6, 0.3–14)	< 0.001	9 (10, 0.3–20)
Involved structures**				
Iris (yes)	3/46	2/32	0.96	5/78
Ciliary body (yes)	29/46	10/32	0.006	39/78
Choroid (yes)	46/46	31/32	0.23	77/78
EOE (yes)	4/47	1/34	0.32	5/81
AJCC (cTNM)				
T category**				
T1 (T1a, T1b, T1c, T1d)	2 (2, 0, 0, 0)	9 (7, 0, 2, 0)	0.004	11 (9, 0, 2, 0)
T2 (T2a, T2b, T2c, T2d)	5 (3, 1, 0, 1)	5 (5, 0, 0, 0)	0.60	10 (8, 1, 0, 1)
T3 (T3a, T3b, T3c, T3d)	11 (7, 3, 0, 1)	14 (8, 6, 0, 0)	0.09	25 (15, 9, 0, 1)
T4 (T4a, T4b, T4c, T4d, T4e)	27 (5, 21, 1, 0, 0)	5 (1, 4, 0, 0, 0)	< 0.001	32 (6, 25, 1, 0, 0)
N category (yes)	0/47	0/34	--	0/81
M category (yes)	0/47	0/34	--	0/81
Reason for enucleation				
Large tumor	27/47 (57%)	0/34 (0%)	< 0.001	27/81 (33%)
High-risk factors	11/47 (23%)	2/34 (6%)	0.03	13/81 (16%)
Recurrence	0/47 (0%)	23/34 (68%)	< 0.001	23/81 (28%)
Blind painful eye	0/47 (0%)	8/34 (23%)	< 0.001	8/81 (10%)
Patient decision	9/47 (19%)	1/34 (3%)	0.02	10/81 (12%)

EOE=extraocular extension; AJCC cTNM=American Joint Committee on Cancer, 8th ed..ition, clinical classification for uveal melanoma; T=tumor; N=lymph nodes; M=metastasis *Two-tailed P **In tumors with visible tumor margins

data and metastases) [Fig. 2c and d]. Seven of 14 (50%) patients with lost nBAP1 were discordant with respect to either genetic status (3 tumors with ch3D) or outcome (4 patients with ch3M or no genetic data and no metastases) [Fig. 3a and b, Table 3].

Association of nBAP1 immunoreactivity with chromosome 3 status and outcome [(ch3D OR no metastasis)/intact nBAP1

AND (ch3M OR metastasis)/lost nBAP1] in post-brachytherapy enucleated uveal melanomas was significantly lower when compared with non-brachytherapy tumors [21/30 (70%) vs. 41/44 (93%), $P = 0.004^*$] [Table 3]. A subset analysis of UM with lost nBAP1 expression demonstrated a continued statistically significant difference in concordance with chromosome 3 status and outcome between the post-brachytherapy and

Table 2: Histopathologic characteristics of uveal melanoma in 81 patients comparing primary versus secondary enucleation

	Primary enucleation (%) Non-irradiated melanoma (n=47 patients, 47 eyes)	Secondary enucleation (%) Irradiated melanoma (n=34 patients, 34 eyes)	P*	All patients (%) (n=81 patients, 81 eyes)
Tumor size (mm)				
Basal diameter (median, range)	14 (15, 4-21)	13 (12, 5-22)	0.24	13 (15, 4-22)
Thickness (median, range)	9 (10, 1-20)	5 (4, 0.5-14)	<0.001	7 (8, 0.5-20)
Involved structures				
Iris (yes)	5/47 (11%)	3/34 (9%)	0.79	8
Ciliary body (yes)	23/47 (49%)	11/34 (32%)	0.14	34
Choroid (yes)	47/47 (100%)	34/34 (100%)	--	81
Sclera (yes)	16/47 (34%)	13/34 (38%)	0.70	29
EOE (yes)	4/47 (8%)	4/34 (12%)	0.63	8
ON (yes)	9/47 (19%)	6/34 (18%)	0.87	15
Vitreous/retina seeding (yes)	12/47 (25%)	9/34 (26%)	0.92	21
Anterior chamber angle (yes)	9/47 (19%)	5/34 (15%)	0.60	14
Growth pattern				
Dome	19/47 (40%)	17/34 (50%)	0.39	36/81 (44%)
Mushroom	23/47 (49%)	8/34 (24%)	0.02	31/81 (38%)
Diffuse	5/47 (11%)	9/34 (26%)	0.63	14/81 (17%)
High risk factors				
Mitotic figures in 40 HPF (<10)	16/47 (34%)	19/32 (59%)	0.03	35
Vacuogenic mimicry (yes)	26/47 (55%)	12/34 (35%)	0.08	38
Vascular invasion (yes)	2/47 (4%)	0/34 (0%)	0.22	2
Lymphocytes (brisk)	21/47 (45%)	12/34 (35%)	0.39	33
Melanophages (brisk)	20/47 (42%)	15/34 (44%)	0.89	35
Necrosis (yes)	13/47 (28%)	23/34 (68%)	<0.001	36
AJCC (pTNM)				
T category				
T1 (T1a, T1b, T1c, T1d)	8 (6, 1, 0, 1)	12 (9, 2, 1, 0)	0.60	20 (15, 3, 1, 1)
T2 (T2a, T2b, T2c, T2d)	3 (2, 1, 0, 0)	7 (5, 2, 0, 0)	0.55	10 (7, 3, 0, 0)
T3 (T3a, T3b, T3c, T3d)	29 (14, 12, 2, 1)	9 (3, 5, 1, 0)	0.002	38 (17, 17, 3, 1)
T4 (T4a, T4b, T4c, T4d, T4e)	7 (1, 6, 0, 0, 0)	6 (1, 2, 0, 0, 3)	0.74	13 (2, 8, 0, 0, 3)
N category (N0)	47/47	33/34	0.24	80/81
M category (M0)	47/47	31/34	0.04	78/81
G category [spindle, non-spindle (mixed, epithelioid)]	13, 34 (32, 2)	5, 29 (26, 3)	0.17	18, 63 (58, 5)

EOE=extraocular extension; ON=optic nerve; AJCC pTNM=American Joint Committee on Cancer, 8th ed..ition, pathologic classification for uveal melanoma;

T=tumor; N=lymph nodes; M=metastasis; G=grade *Two-tailed P

non-brachytherapy tumors [10/14 (71%) vs. 24/24 (100%), $P = 0.003^*$] [Table 3]. There was no significant difference in concordance with chromosome 3 status and outcome between the postbrachytherapy and nonbrachytherapy UM with intact nBAP1 expression [11/16 (69%) vs. 17/20 (85%), $P = 0.12$].

During our initial evaluation of adequacy of BAP1 IHC we excluded all cases with absent BAP1 staining of the internal positive controls, the retina and/or intratumoral blood vessels and inflammatory cells. On re-evaluation of discordant cases, six postbrachytherapy UMs with lost nBAP1 expression demonstrated weak intratumoral internal positive control (decreased nuclear staining in vascular endothelium and inflammatory cells) and either weak or absent staining of the retina immediately overlying the tumor [Fig. 3c and d], suggestive of a falsely negative nBAP1 expression. With those cases excluded from the analysis, the association of nBAP1

immunoreactivity with chromosome 3 status and outcome was still significantly lower in postbrachytherapy compared to nonbrachytherapy tumors [18/24 (75%) vs 41/44 (93%), $P = 0.02^*$] [Table 3].

Discussion

The loss of the nBAP1 expression by IHC in UM correlates with genetic prognostic markers including somatic *BAP1* gene mutation,^[5] chromosome 3 monosomy,^[6] and GEP class 2.^[8] Several studies have documented that BAP1 IHC is an independent prognostic factor of metastatic-free survival,^[7] possibly superior to somatic *BAP1* gene mutation status.^[5,9]

In the current study, we compared the reliability of the BAP1 immunohistochemical stain in prognostication of

Table 3: Immunohistochemistry and molecular genetic features of uveal melanoma in 81 patients comparing primary versus secondary enucleation

	Primary enucleation (%) Non-irradiated melanoma (n=47 patients, 47 eyes)	Secondary enucleation (%) Irradiated melanoma (n=34 patients, 34 eyes)	P*	Secondary enucleation (%) Irradiated melanoma (n=28 patients, 28 eyes)*****	P*
Tumor harvesting/FNAB route					
Trans scleral	0/47 (0%)	7/22 (32%)	<0.001	5/18 (28%)	<0.001
Pars Plana	0/47 (0%)	10/22 (45%)		8/18 (44%)	
On enucleated globe	47/47 (100%)	5/22 (23%)		5/18 (28%)	
Not done	0/47 (0%)	12/34 (35%)		10/28 (36%)	
nBAP1 IHC					
Intact	20/47 (43%)	16/34 (47%)	0.34	16/28 (57%)	0.11
Lost	24/47 (51%)	14/34 (41%)	0.19	8/28 (29%)	0.03
Non-interpretable**	3/47 (6%)	4/34 (12%)	0.20	4/28 (14%)	0.13
BAP1 mutation status					
Wild-type	2/5 (40%)	1/3 (33%)	0.42	1/3 (33%)	0.42
Somatic mutation	3/5 (60%)	1/3 (33%)	0.23	1/3 (33%)	0.23
Germline mutation	0/5 (0%)	1/3 (33%)	0.08	1/3 (33%)	0.08
Not known	42/47 (89%)	31/34 (91%)	0.39	25/28 (89%)	0.50
Chromosome 3 status					
Disomy***	19/47 (40%)	9/22 (41%)	0.48	7/17 (41%)	0.48
Monosomy	28/47 (60%)	13/22 (59%)	0.48	10/17 (59%)	0.48
Not known	0/47 (0%)	12/34 (35%)	<0.001	11/28 (39%)	<0.001
Association of nBAP1 IHC, Chromosome 3 status and outcome					
Intact nBAP1					
ch3D***/Intact nBAP1	16/20 (80%)	5/8 (63%)	0.17	5/8 (63%)	0.17
No mets****/Intact nBAP1	9/11 (82%)	9/15 (60%)	0.18	9/15 (60%)	0.18
ch3D*** AND no mets****/Intact nBAP1	8/11 (73%)	3/7 (43%)	0.10	3/7 (43%)	0.10
ch3D*** OR no mets****/Intact nBAP1	17/20 (85%)	11/16 (69%)	0.12	11/16 (69%)	0.12
Lost nBAP1					
ch3M/Lost nBAP1	24/24 (100%)	7/10 (70%)	0.002	4/5 (80%)	0.01
mets****/Lost nBAP1	13/19 (68%)	7/13 (54%)	0.20	7/7 (100%)	0.045
ch3M AND mets****/Lost nBAP1	13/19 (68%)	4/9 (44%)	0.11	4/4 (100%)	0.095
ch3M OR mets****/Lost nBAP1	24/24 (100%)	10/14 (71%)	0.003	7/8 (88%)	0.04
ch3D*** OR no mets****/Intact nBAP1 and ch3M OR mets****/Lost nBAP1	41/44 (93%)	21/30 (70%)	0.004	18/24 (75%)	0.02

*One-tailed P. **No internal control in the immunostained slide or <5% viable tumor. ***One case with Chromosome 3q amplification is included in this group.

****In patients with >60 months follow-up or metastatic disease development. *****Removing 6 cases with lacking internal control in re-review. nBAP-1 IHC=nuclear BRCA-associated protein-1 immunohistochemical stain; FNAB=fine needle aspiration biopsy; ch3=chromosome 3, ch3M=chromosome 3 monosomy, ch3D=chromosome 3 disomy, mets=metastasis

patients who underwent secondary enucleation post I-125 brachytherapy for UM with patients who underwent primary enucleation. We confirmed an association of nBAP1 expression with chromosome 3 status and outcome in non-brachytherapy UM comparable to prior studies, confirming its role in the prognostication of patients who underwent primary enucleation.^[5-9] However, we noted a significantly lower association between nBAP1 expression and these parameters in post-brachytherapy UM. The discordance was particularly significant in the post-brachytherapy tumors with lost nBAP1 expression. Re-evaluation of these discordant cases demonstrated a weak internal nBAP1 control, suggesting that radiation-induced changes in tissue antigenicity may have

resulted in a false-negative nBAP1 expression. However, even with exclusion of these cases from analysis, the association between nBAP1 expression, chromosome status, and outcome in post-brachytherapy UM was still significantly lower when compared with nonbrachytherapy tumors ($P = 0.02^*$), limiting nBAP1 stain role in prognostication of patients who underwent postbrachytherapy secondary enucleation. In addition to the alterations in tissue antigenicity, irradiation-induced epigenetic effects may have contributed to the discrepancy between nBAP1 expression and chromosomal status. Our findings are in keeping with observations by Szalai *et al.*,^[21] who noted a significant difference in BAP1 expression between the post-brachytherapy and non-brachytherapy UM, although the

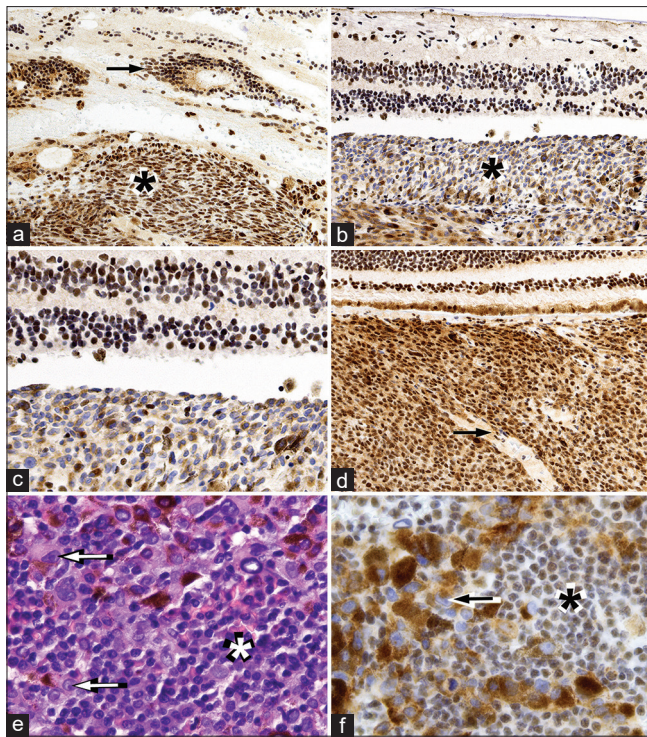


Figure 1: Primary and postbrachytherapy uveal melanomas with concordant nuclear BAP1 (nBAP1) expression, chromosome 3, and outcome status. (a) Intact nBAP1 staining in most retinal cells (arrow) and the underlying spindle cell type uveal melanoma (asterisk) with disomy 3. (b) Lost nBAP1 staining in the uveal melanoma (asterisk) with monosomy 3. (c) Higher magnification of tumor and overlying retina in (b). (d) Intact nBAP1 staining in post-brachytherapy uveal melanoma with disomy 3, in intratumoral blood vessels (arrow), and the overlying retina. (e) Postbrachytherapy epithelioid cell type (arrows) uveal melanoma that metastasized demonstrates intense intratumoral lymphocytic infiltrate (asterisk). (f) The tumor in (e) demonstrates loss of nBAP1 expression (arrow) with retained nBAP1 expression in tumor-infiltrating lymphocytes (asterisk). [BAP1 stain (a, b, c, d, f), hematoxylin-eosin stain (e); x200 (a and b), x400 (c), and x650 (e and f)]

authors did not provide specific information on the staining patterns between the two groups.

Our study has several important limitations. Although prior studies demonstrated no appreciable differences in the genetic profiles of nonbrachytherapy versus postbrachytherapy UM and in FNAB-based versus tissue-based profiling,^[14-16] our results are limited by the absence of genetic studies on paraffin-embedded tissue. FNAB was performed for genetic profiling only and the material was not available for cytopathologic and immunohistochemical evaluation. The study is also limited by the incomplete outcome data. A significant drawback of our study is limited *BAP1* mutation data for correlation with BAP1 IHC status. Additionally, we do not have the genetic data on other prognostically important genes, including *SF3B1* and *EIF1AX*.^[23] The incomplete chromosome 3 and 8 data on postbrachytherapy tumors may have introduced a bias in our evaluation of two study groups. The non-brachytherapy tumors were more advanced with more frequent high-risk clinical and histopathologic features when compared to the post-brachytherapy tumors.

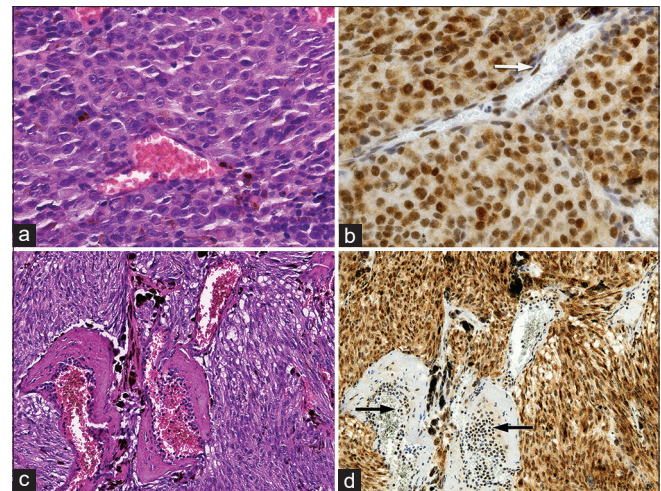


Figure 2: Primary and post-brachytherapy uveal melanomas with discordant intact nuclear BAP1 (nBAP1) expression and either monosomy 3 or metastasis status. (a) Mixed cell type primary uveal melanoma with monosomy 3. (b) The tumor in (a) demonstrates intact nBAP1 expression and a patchy (wild-type) nBAP1 expression in the vascular endothelium of intratumoral vessels (arrow). (c) Spindle cell type irradiated uveal melanoma with monosomy 3. (d) The tumor demonstrates intact nuclear BAP1 expression. A patchy (wild-type) nBAP1 expression is noted in the vascular endothelium and inflammatory cells (arrows). [hematoxylin-eosin stain (a and c), BAP1 stain (b and d); x400 (a and b), and x100 (c and d)]

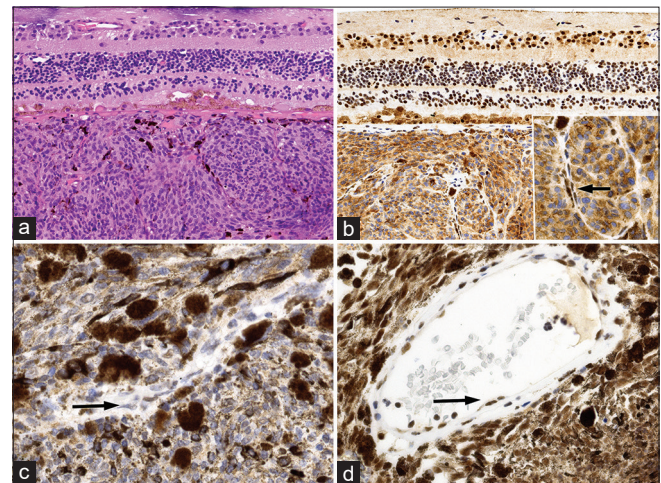


Figure 3: Postbrachytherapy uveal melanoma with discordant lost nuclear BAP1 (nBAP1) expression and disomy 3 status. (a) Mixed cell type uveal melanoma. (b) Diffuse loss of nBAP1 in the tumor. Strong preserved nuclear staining in the overlying retina and the intratumoral blood vessels (inset, arrow). (c) Disomy 3 uveal melanoma demonstrates loss of nBAP1 expression with weak staining in the intratumoral vascular endothelium (arrow). (d) A small focus in the same tumor demonstrates intact nBAP1 expression with strong nuclear staining in the vascular endothelium of the adjacent vessel (arrow). [hematoxylin-eosin stain (a), BAP1 (b-d); x200 (a and b), and x400 (inset b-d)]

Although these differences likely affected genetics and outcome, they should not have influenced the reliability of immunohistochemical stain in each study group. It would be interesting to investigate if dosimetric data on the postbrachytherapy UM may correlate with false negative

Table 4: Follow-up data and date last seen clinical characteristics of uveal melanoma in 81 patients comparing primary versus secondary enucleation

	Primary enucleation (%) Non-irradiated melanoma (n=47 patients, 47 eyes)	Secondary enucleation (%) Irradiated melanoma (n=34 patients, 34 eyes)	P*	All patients (%) (n=81 patients, 81 eyes)
Follow-up duration (months)				
Mean (median, range)	52 (53, 0.1–111)	90 (81, 1–307)	<0.001	68 (69, 0.1–307)
Local recurrence				
Yes	1/30 (3%)	2/30 (7%)	0.55	3/60 (5%)
Prophylactic Sunitinib (Stent)				
Yes (mean duration)	8 (6 months)	1 (2 months)	0.047	9 (5 months)
Systemic metastatic disease				
No metastasis	17/32 (53%)	18/32 (56%)	0.80	35/64 (55%)
Metastasis	15/32 (47%)	14/32 (44%)	0.80	29/64 (45%)
Unknown**	15/47 (32%)	2/34 (6%)	0.005	17/81 (21%)
Outcome				
No metastasis	16/32 (50%)	18/32 (56%)	0.62	34/64 (53%)
Alive with metastasis	5/32 (16%)	4/32 (13%)	0.72	9/64 (14%)
Dead from disease	10/32 (31%)	10/32 (31%)	1	20/64 (31%)
Dead no metastasis	1/32 (3%)	0/32 (0%)	0.31	1/64 (2%)
Unknown**	15/47 (32%)	2/34 (6%)	0.005	17/81 (21%)

*Two-tailed P **Patients with <60 months follow-up

BAP1 IHC. Unfortunately, the dosimetric data in our study group is not available for this analysis. Finally, although the relatively small number of cases in our study prevent detailed multivariate analyses, we believe our findings highlight the key potential pitfalls in the interpretation of this important UM biomarker.

Conclusions

In conclusion, our study demonstrates a significantly lower association between nBAP1 expression, chromosome 3 status, and outcome in postbrachytherapy UM when compared to non-brachytherapy tumors. These observations highlight some of the potential limitations of BAP1 immunohistochemistry in the prognostication of postbrachytherapy UM patients. The absence of nBAP1 expression should be interpreted with caution, particularly when no strong internal control is present. Future studies that incorporate genetic data are needed to more comprehensively assess the prognostic reliability of nBAP1 expression in irradiated UM.

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Institutional review board statement

The study was conducted in accordance with the Declaration of Helsinki and approved by the Institutional Review Board of the Wills Eye Hospital (Protocol code IRB #2020-35 and date of approval 4/22/2020).

Informed consent statement

Patient consent was waived due to the retrospective nature of the study.

Simple Summary: The BRCA-associated protein1 (BAP1) immunohistochemical (IHC) stain has emerged as a powerful and inexpensive prognostic tool in uveal melanoma (UM), correlating with molecular genetics and patient outcome. The reliability of BAP1 IHC in post-Iodine-125 plaque brachytherapy-treated UM-enucleated eyes has not been formally assessed. In this study, 34 postbrachytherapy-enucleated eyes with UM were compared to 47 nonbrachytherapy controls. There was no significant difference between postbrachytherapy and nonbrachytherapy UM with respect to BAP1 IHC, chromosome 3 status, and outcome. Association of BAP1 IHC with chromosome 3 status and outcome in postbrachytherapy UM was significantly lower when compared with nonbrachytherapy tumors.

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Conflicts of interest: There are no conflicts of interest.

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